

WASP-135b

R-0222

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_210731 · created 2026-07-17T01:51:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459426.7215 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.05
Transit depth (Rp/Rs)^2	7.8e-07 +/- 0.00088 [%]
Orbital Inclination (inc)	83.01 +/- 2.95
Airmass coefficient 1 (a1)	17.84 +/- 10.09
Airmass coefficient 2 (a2)	-3.0 +/- 1.44
Scatter in the residuals of the lightcurve fit is	49.75 %
Best Comparison Star	#2 - [58, 127]
Optimal Aperture	41.83
Optimal Annulus	132.45
Transit Duration (day)	0.053 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 ± 0.05 vs 0.139 (deviation 2.3σ)
Duration fit vs published	0.053 d vs 0.0687 d (deviation 0.59σ)
Mid-time O–C	5.3 min
Depth SNR	0.0
Residual scatter	49.75 %

Light curve

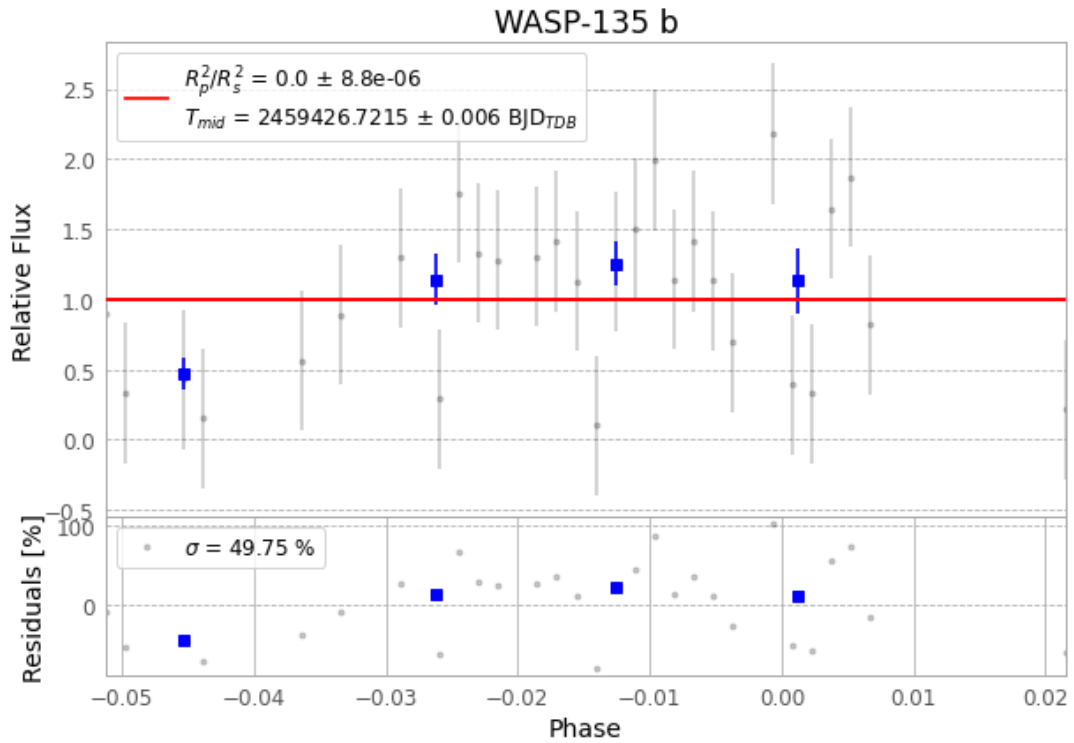


Figure 1 — FinalLightCurve_WASP-135 b_2021-07-30.png (SHA-256 ad686f3a9800cb74...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [106, 277], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e84b87bf85c3cadb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

52 FITS frames (34 MB), WASP-135210731032415.FITS → WASP-135210731055712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-210731.log (0790b8a25d45...), FinalLightCurve_WASP-135 b_2021-07-30.png (ad686f3a9800...), FinalLightCurve_WASP-135 b_2021-07-30.csv (d644f365686f...)

Compute resources

Wall time 325.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 01:51Z.